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Research on identity document image tampering detection based on texture understanding and multistream networks

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ABSTRACT. Identity document images are widely utilized in personal identity authentication systems. However, the rapid development of image-editing tools poses significant risks to personal privacy and information security. Consequently, tampering detection in identity document images has become a crucial research focus. The current challenges in this field primarily stem from two aspects: the increasing realism of tampered content and the prevalence of secondary editing operations on manipulated images. We propose a multistream network architecture that leverages the distinctive background texture patterns of identity documents. This model enables simultaneous extraction and effective fusion of three critical feature domains: background texture characteristics, frequency-domain representations, and noise distribution. To support rigorous evaluation, we specifically develop the tampered-identity dataset containing primarily secondarily edited tampered samples. Experiments conducted on the tampered-identity dataset as well as other publicly available datasets demonstrate that the proposed method exhibits commendable detection performance, particularly in effectively identifying tampering that has been formed through secondary editing.

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1 Introduction

In the information age, digital platforms spanning financial technology,[1] e-commerce,[2] and audit certification systems[3] are undergoing unprecedented refinement. They are deeply integrated into daily life and are revolutionizing our interaction with essential services. This digital transformation empowers users with lightning-fast transactions, frictionless verification processes, and enterprise-grade security protocols, reshaping modern civilization with efficiency.

Central to this evolution is the exponential growth of digitally authenticated document imaging, which includes government-issued IDs, electronic financial records, legally binding contracts, and privacy-sensitive judicial materials. These images are far beyond mere digital replicas and serve as digital fingerprints. They are directly linked to the legal validity of transactions, institutional credibility, and citizens' fundamental rights. Among them, identity document images are the cornerstones of civil infrastructure. They are irreplaceable in scenarios such as university admissions, career advancement, and cross-border legal procedures, with each pixel carrying the weight of personal sovereignty.

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Fig. 1 Examples of identity document image tampering types, featuring a comparison between original and tampered images, as well as the details of the tampering.

In this era of rapid technological advancement, image manipulation tools have achieved unprecedented sophistication. Malicious actors now leverage AI-empowered editing tools to surgically alter identity document images. Such actions trigger large-scale threats to public safety, including identity theft, counterfeit academic certifications, and cross-border financial scams. These digitally engineered crises not only undermine civic trust but also cause systemic vulnerabilities, eroding governmental credibility. Within judicial ecosystems, doctored evidentiary images jeopardize judicial integrity, corroding the bedrock principles of due process and democratic rule of law. Thus, advancing tamper-detection algorithms not only safeguards information authenticity and cybersecurity sovereignty but also ultimately preserves the social contract itself.

As shown in Fig. 1, for identity document images, the four main types of tampering currently prevalent are as follows:

(1) Copy-Move: This involves duplicating specific regions of an image and relocating them to other areas within the same image, thereby obscuring or replicating particular sections.
(2) Splice: This technique merges elements from disparate images into a unified composition.
(3) Erase: Utilizing erasing tools to perform localized edits, such as excising or masking segments of an image. Examples include blemish removal, color modification, and detail enhancement or reduction.
(4) Generation: Crafting new content that closely matches the surrounding style in areas that have been erased or in blank regions, making it imperceptible to the naked eye.

Currently, the definition of tampering types tends to classify types (3) and (4) as a single tampering form. A number of methods[2,3] use locating text positions as a preliminary step for image tampering detection. However, because there is no text in the erased area, this approach is not conducive to learning tampering features. Consequently, we consider them as two distinct types of tampering.

Currently, the focus of image tampering detection technology is primarily on the domain of natural images. However, there are distinct differences between image tampering and identity document image tampering, which are document-based images:

(1) Size of the tampered area: The tampering in natural images mainly targets certain objects within the image, resulting in larger tampered regions. By contrast, the tampering in

identity document images primarily focuses on the textual content, leading to smaller tampered areas.

(2) Perceptibility of tampering traces: Identity document images typically have a relatively uniform background with strong texture, and the font size and style of the text are consistent. It is difficult to capture the differences in local texture, and the local texture differences between tampered and genuine text are very vague. On the other hand, natural images have complex backgrounds, and the differences in details such as the edges and colors of different objects within the image are more apparent.

(3) Factors interfering with tampering detection: Identity document images are usually obtained through photographing or scanning real documents, or by photographing the scanned copies, and are affected by factors such as paper wrinkles, light shadows, and geometric distortions caused by lens differences, which impact the detection performance. By contrast, the interference information in naturally tampered images mainly comes from camera noise or noise generated during post-processing.

Therefore, it is not feasible to directly apply natural image tampering detection technology to the task of tampering detection in identity document images and other document-based images.

Currently, the task of identity document image tampering detection confronts a multitude of significant challenges:

1. The high similarity in texture between genuine and tampered text makes it difficult to accurately localize tampered regions through feature learning based solely on a single-domain perspective (e.g., the RGB domain). Considering the distinct background texture of identity document images and the discontinuity of tampered regions in the frequency domain, these characteristics can be utilized to assist the network in differentiating between authentic and tampered areas. Moreover, noise features have the ability to highlight the key content within an image while suppressing irrelevant information.
2. Unlawful individuals may utilize various techniques to perform secondary editing on images that have already been tampered with, with the aim of concealing the traces of tampering. This can interfere with the detection model's analysis, thereby reducing its accuracy and reliability in identifying tampered images. A common example is image compression. Image compression possesses distinct characteristics that, under certain circumstances, may impact the assessment of image integrity. For instance, the compression process may alter the details and textures of an image to a certain extent, thus masking tampering traces that might have originally been detectable.

Therefore, we propose an identity document image tampering detection method based on texture understanding and a multistream network. It fully leverages the RGB, texture, frequency domain, and noise aspects of images. We designed a parallel three-branch multifeature network. It can enhance the discriminative ability for tampering textures from multiple perspectives, thoroughly explore tampering features, and create a more comprehensive representation of tampering information. In addition, to verify the performance of our method and enhance the model's robustness to secondary editing, we introduced a new identity document image tampering dataset named Tamper-Identity. This dataset comprises 900 ID tampering images of various types, covering a range of mainstream tampering types.

The principal contributions of this paper are presented below:

(1) We have constructed a multistream tampering detection network for identity document images, which comprehensively and thoroughly explores tampering features from various perspectives.

(2) A multistream feature perceptron has been designed, which can fully perceive different tampering features of an image from a multidomain perspective, and fully extract the texture information, RGB information, frequency-domain information, and noise information of tampered images.

(3) A multiscale feature fusion structure was designed for different perspective features to achieve feature interaction at different scales, thereby enhancing the key content

information in tampered images and improving the accuracy of locating tampered regions in the identity document image.

(4) A dataset encompassing various types of identity document images and tampering types has been created, which comprises both manual and automatic tampering methods performed using diverse techniques. The images have undergone secondary editing to enhance the quality of the tampering.

2 Related Works

2.1 Classification of Image Tampering Detection Methods

From the perspective of whether information is embedded in advance, tampering detection can be categorized into active detection and passive detection. Active detection techniques, such as digital watermarks and digital signatures,[1,4,5] necessitate the embedding of specific identification information at the initial stages of image creation or distribution, providing direct evidence for subsequent tampering detection. For instance, digital watermarking technology embeds imperceptible marks into image data, which does not affect the visual quality of the image and can effectively carry key information such as copyright, identity authentication, or integrity check codes. However, these methods require the cooperation of the image provider, which is often difficult to achieve in practice, making passive detection technology a research hotspot.

Passive detection technology does not depend on pre-embedded information, but rather identifies tampering traces by analyzing the inherent features of the image itself. For instance, tampering detection can be implemented by exploiting the characteristics of different data storage formats. Taking JPEG images as an example, by utilizing their lossy compression[6,7] feature, the double JPEG compression detection method[8] can be employed to detect tampering in JPEG-formatted images.

From the perspective of the features utilized, passive detection technology can be categorized into methods based on traditional features and methods based on deep learning features. Methods based on traditional features primarily utilize predesigned feature operators to detect inconsistencies within images, such as detection methods based on overlapping blocks, feature points, image attributes, device attributes, and compression attributes. These methods, through in-depth analysis of the natural features of images, are capable of detecting tampering traces; however, they have relatively high technical complexity. With the improvement of computer performance and the widespread application of deep learning methods in the field of computer vision, researchers have begun to widely apply neural networks to tampering detection tasks.

2.2 Tampering Detection Methods in Deep Learning

As image forgery technology advances, traditional tampering detection methods fall short in meeting visual recognition requirements. Deep-learning–based detection technology has become the main solution. Current mainstream methods mainly focus on two key aspects: the diversity of feature representation and the effectiveness of feature fusion.

Basic image features involve RGB, noise, artifact, frequency-domain, and texture features, which can reflect potential tampering traces in images from different angles. Compared with Rao et al.,[9] single RGB feature extraction and learning have limited representational capability for tampering traces and marked limitations in capturing the high-frequency ones left by tampering operations. At present, the most frequently used features are those based on the frequency domain representations and noise distribution. Wang et al.[2] first introduced frequency-domain features to enhance tampering texture feature detection in document image tampering detection tasks. By capturing both RGB and frequency-domain features, the network's ability to distinguish local texture differences was improved. Wu et al.[10] simulated noise generated during social network transmission, trained networks with modeled noise, and enhanced the robustness of image tampering detection methods. Chen et al.[11] proposed MVSS-Net, which extracts noise and boundary artifact features through multiview and multiscale supervision to learn tampering detection features. Liao et al.[12] constructed a character texture perception network to capture text-region features in document images. Although it performs well in post-processed document images, its effectiveness in detecting tampering traces caused by erasing is relatively weak. Liu et al.[13] proposed the dual-path progressive structure PSCC-Net, which uses a nonlocal

attention mechanism to capture global context relationships in feature maps, effectively identifying semantic inconsistencies caused by splicing, copy-move operations, etc. Recently, integrating multimodalities (including image and text modalities) into image tampering detection has become a new trend. Xu et al.[14] and Sun et al.[15] used multimodal large language models, combining visual and textual information for tampering detection. This method not only improves the effectiveness of image forgery detection and localization but also provides interpretable evidence, offering a more comprehensive and explainable solution to the challenges in image tampering detection. However, detection methods combining large language models consume a lot of computational resources.

When performing feature fusion, common methods currently include feature pyramids[16] and attention-guided fusion.[17,18] Feature pyramids have the advantage of enabling multiscale feature fusion, but they have limited positioning accuracy and high computational complexity. Attention-guided fusion not only can adaptively adjust the importance of features but also has high computational complexity.

Despite the significant progress achieved by existing methods in image tampering detection and localization, we have found that, owing to the continuous evolution of tampering techniques, current deep learning-based image tampering detection methods still have limitations. Moreover, the variety of tampering methods and the differences between images from diverse sources can also influence the recognition results. At present, there is a notable scarcity of datasets that encompass multiple tampering types, and models trained solely on a single-tampering-type training set have limited generalization ability. The question of how to expand tampered image datasets while more effectively utilizing diverse features to strengthen the robustness of models requires further in-depth investigation.

3 Method

The identity document image tampering detection method we proposed, which is based on texture understanding and a multistream network, is composed of three parallel network branches and a final multiscale feature fusion module. The three parallel branches consist of the texture feature branch, the frequency-domain feature branch, and the noise feature branch, each of which includes a corresponding submodule of the multistream feature perceiver. This detection method fully mines the tampering information of identity document images from multiple feature perspectives, obtains a more comprehensive and integrated representation of tampering features, and enhances the comprehensive perception of identity document image tampering. The overall network model framework is illustrated in Fig. 2. The identity document image to be detected, $I_{\text{ori}} \in R^{3\times H\times W}$, is input into the multistream feature perceiver submodules of the three parallel texture feature branch, frequency-domain feature branch, and noise feature branch. These submodules transform the image into feature representations from different perspectives, F_{texture}, $F_{\text{frequency}}$, and F_{noise}, respectively. Subsequently, the feature representations are input into the backbone network of each branch for feature extraction. Finally, the feature representations D_{tm}, D_{fn}, and D_{ni} extracted by each branch are input into the multiscale feature fusion module for feature fusion to obtain the final feature representation.

3.1 Multi-stream Feature Perceptron

The multistream feature perceptron, functioning as the head module of each branch, is capable of perceiving features from different perspectives. It transforms the original input ID tampering images into feature representations that can be input into each branch. This perceiver comprises three submodules: the texture enhancement module, the JPEG artifact learning module,[19] and BayarConv,[20] each of which is used for the input feature transformation of the corresponding branch. We will discuss the three modules in detail in the corresponding branch sections below.

3.1.1 *Texture feature branch*

Different image tampering methods involve modifications to pixels, resulting in inconsistencies in the color and texture of the identity document image. This branch is based on texture and RGB features for feature extraction. As shown in Fig. 3 (left), various types of IDs typically have line textures added to the background area for the purposes of anticounterfeiting and aesthetics during

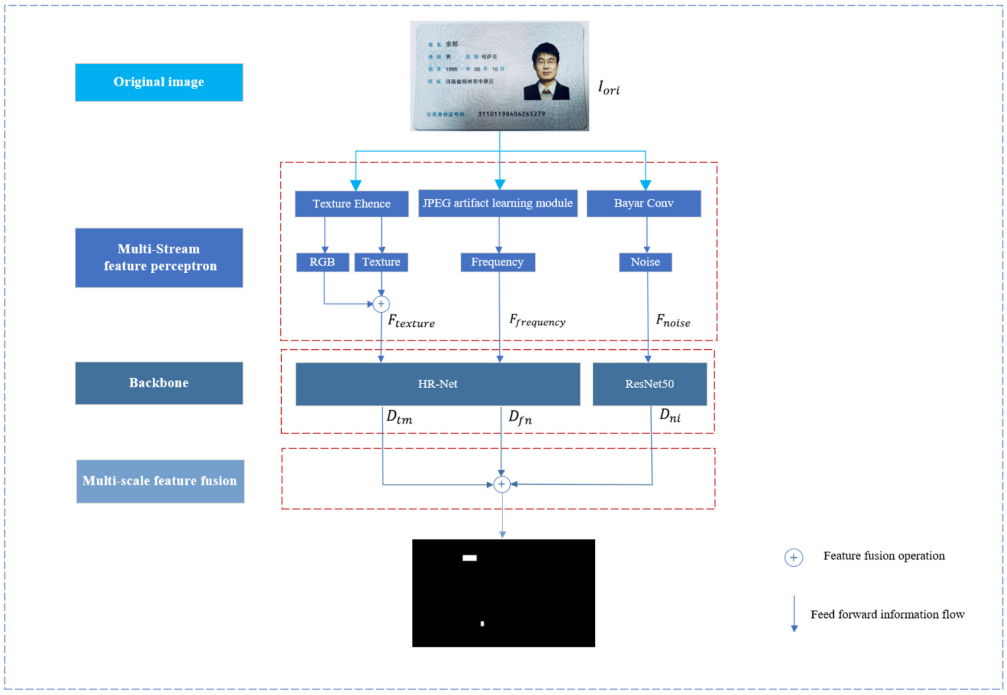


Fig. 2 Overall architecture of our model. We employ a multistream feature perceptron to transform the input image into features to be extracted from different perspectives. These features are then fed into the backbone network for extraction. Finally, feature fusion is carried out via a multiscale feature fusion module.

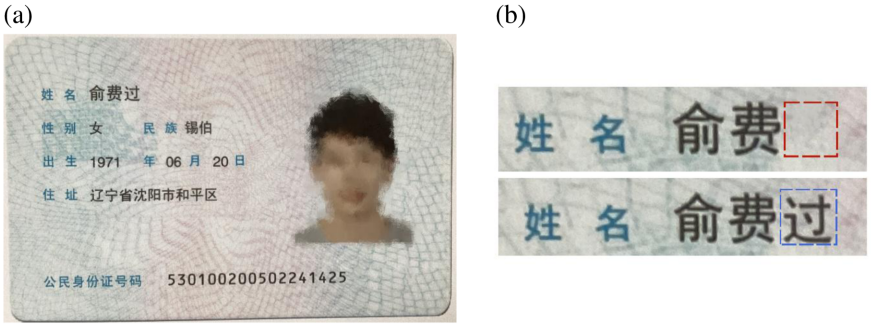


Fig. 3 Global texture features of identity document image (a) and the impact of tampering operations on background texture (b). The region within the blue box is the pretampering area, and the region within the red box is the post-tampering area.

their design and production. These texture patterns usually exhibit the characteristics of being globally regular and locally random.

Various image forgery methods involve adjustments at the pixel level, which can lead to inconsistencies in color and texture in identity document images. As depicted in Fig. 3 (right), the texture features of the tampered regions become more blurred and disordered. Leveraging this characteristic of identity document images, we designed a texture enhancement module to enhance the representation of image texture features.

Most binary classification methods for image tampering detection have overlooked an important phenomenon, namely, that forgeries caused by forgery methods are often evident in the texture information of shallow feature maps. The texture information here represents the high-frequency components of shallow features. Therefore, to retain more texture information to capture these artifacts, we designed the texture feature enhancement module as shown in Fig. 4. First, the input image is passed through two convolution operations to obtain the shallow feature map L_1. Subsequently, intrablock local average pooling is applied to downsample L_1, resulting in

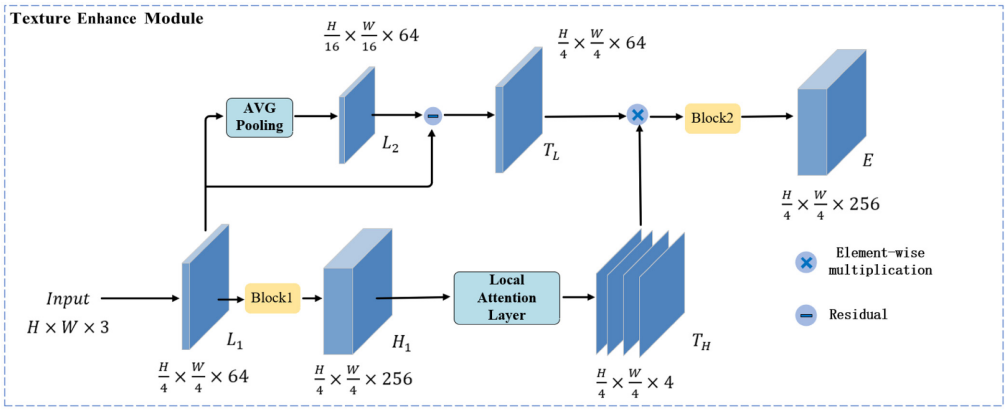


Fig. 4 Texture enhancement module structure.

the pooled feature map L_2. Then, similar to the texture representation of spatial images, we defined the residual of the feature layer to represent the global texture information:

$$T_L = L_1 - L_2. \tag{1}$$

Here, T_L serves as the global texture feature map, which has already incorporated the majority of the global texture information of L_1.

In addition, given that the background texture of some identity document images exhibits local randomness, we designed a local attention layer, which is capable of generating attention maps for different regions. To enable the attention layer to focus on various regions of the input image, guidance from high-level semantic information is required. This necessitates the use of deep-level features from the network. In the texture enhancement module, the shallow feature L_1 is transformed into a deeper feature map H_1 through block 1. Here, block 1 consists of stacked bottlenecks that only adjust the number of channels without changing the size of the feature layer. We use H_1 as the guidance for high-level semantic information and input it into the attention layer to generate attention maps for local regions of the identity document image, denoted as T_H, where $T_H \in R^{\frac{H}{4}\times\frac{W}{4}}$ represents the Hth attention map. As shown in Fig. 5, the attention layer is a lightweight weighted model composed of a 1×1 convolutional layer, a batch normalization layer, an Elu nonlinear activation function, and a predefined mask image M.

In this setup, $M \in R^{24\times24}$ is initialized as a tensor filled with zeros, and the values in the central region of M (excluding a two-pixel-wide border) are set to 1, resulting in a mask that highlights the center.

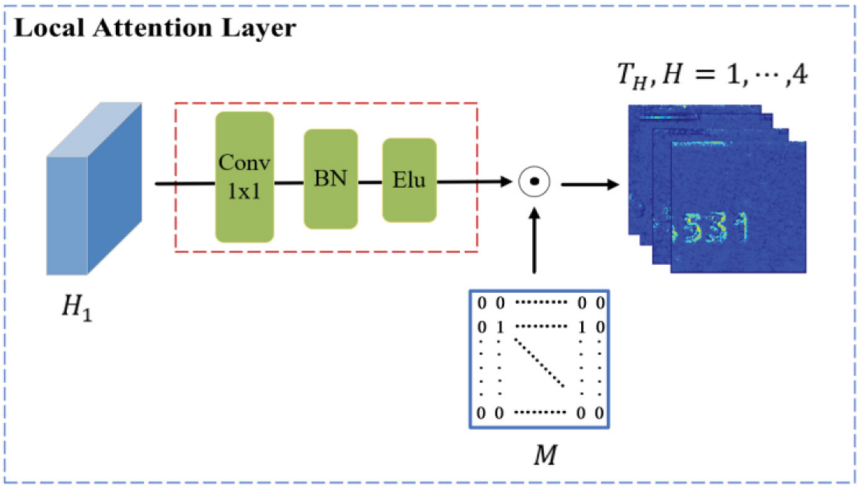


Fig. 5 Structure of local attention layer. The local attention layer structure exploits the globally regular and locally random characteristics of identity document image background textures to effectively acquire their overall background texture features.

In the regional attention layer, a predefined mask M is utilized, and its size is adjusted through interpolation to match the size of the input feature map H_1. M is multiplied by the attention weights, which are generated by a 1×1 convolution and an activation function, and then incremented by 1, to achieve weighted feature maps and generate the final attention map T_H. This process can be expressed as

$$A(X) = \mathrm{ELU}(\mathrm{BN}(\mathrm{Conv}_{1\times1}(X))) + 1, \tag{2}$$

$$M_1 = \mathrm{Interpolate}(M, \mathrm{size} = (H_{H_1}, W_{H_1})), \tag{3}$$

$$T_H = A(X) * M_1. \tag{4}$$

Here, $A(X)$ denotes the attention weights, M_1 represents the mask image after interpolation, Interpolate$(\cdot)$ denotes the interpolation operation, and $*$ indicates weighted multiplication. The mask M is not updated during the training process. This implies that employing the mask M is a flexible approach to controlling the distribution of attention. By adjusting the values of the mask M, the model can be guided to focus on different regions of the feature map. After obtaining the local regional attention map T_H, we perform an element-wise multiplication of the global texture feature map T_L and T_H, combining the low-level texture features and high-level semantic features to form a comprehensive representation for each local region. Subsequently, the multiplied features are enhanced through a three-layer densely connected convolutional block (block 2), and the enhanced features obtained are denoted as E. The above process can be expressed as

$$E = B2(T_L \times T_H). \tag{5}$$

Here, $B2(\cdot)$ denotes feature enhancement using block 2, and $\times$ represents element-wise multiplication. In addition to the aforementioned texture features, we directly extract visual-perception–based features from the input RGB-formatted identity document images. This enables us to identify the visual differences between the tampered and untampered regions and combine these differences with the texture features, thereby enhancing the accuracy of locating the tampered regions.

3.1.2 *Frequency-domain feature branch*

In the frequency-domain feature branch, we focus on compression artifacts. After undergoing DCT transformation, the image is transferred from the RGB domain to the frequency domain, and the resulting frequency components are represented as a set of DCT coefficients. Subsequently, quantization is applied to reduce the spatial occupancy of the image. Following quantization, the DCT coefficients of the image are compressed and encoded to reduce the demand for storage space and transmission bandwidth. The most common compression method currently is the JPEG compression algorithm, which can preserve the visual quality of the image to a certain extent but will generate some compression artifacts.

In the frequency-domain branch, the JPEG artifact learning module is employed to perceive features. This module extracts DCT coefficients and quantization tables from the input image and subsequently feeds them into the feature extraction network. Unlike the traditional DCT histogram, this binary volume representation method enables the network to learn the relationships between adjacent DCT coefficients. The DCT histogram typically merges information on a block-by-block basis, whereas this method preserves image resolution, allowing the network to more finely capture and identify JPEG compression artifacts. In addition, this binary volume representation method is particularly suitable for image segmentation tasks. Because it maintains image resolution and enables the network to learn the relationships between adjacent DCT coefficients, it can be more effectively used for localizing tampered regions in images.

3.1.3 *Noise feature branch*

In real images, noise typically exhibits global consistency, with its distribution and characteristics being closely related to the natural image formation process. This global consistency reflects the random interference that the image experiences during acquisition, transmission, or storage and is an important manifestation of image authenticity. However, when an image is tampered with, the noise distribution in the tampered region often deviates from the global noise pattern of the

original image. This is because tampering operations may introduce new noise sources or alter the noise characteristics of local regions, thereby disrupting the global consistency of the noise.

Unlike conventional approaches, we did not initialize the convolutional layer with the high-pass filter kernels of SRM. Instead, we utilized BayarConv for noise analysis, which has been shown to outperform SRM filters in research.[21] BayarConv is composed of a $5 \times 5 \times 3$ convolutional kernel. In each training iteration, the weight at the center of the convolutional kernel is set to -1, and then, the remaining weights are normalized so that their sum equals 1. This convolutional layer enhances the noise inconsistency between the tampered and genuine regions in the input image, suppresses the pixel information of the original image, and amplifies the noise information. Moreover, the feature extraction network in this branch did not employ pooling layers. Instead, it used convolutions with a stride of 2 to downsample the feature maps. Research has indicated[22] that pooling layers can lead to the loss of noise information, which is undesirable because noise is the most critical feature in this branch.

3.2 Backbone

The texture feature branch and the frequency-domain feature branch utilize HR-Net[23] as the backbone network. Originally designed for computer vision tasks, HR-Net is capable of maintaining high-resolution representations throughout the process, making it suitable for visual tasks that require high-resolution feature representations. Hence, we introduced it to the image tampering detection problem. By contrast, the noise feature branch employs ResNet50 as the backbone network.

Furthermore, as introduced in Sec. 3.1.1 for the texture feature branch, we fused the texture features with the RGB features. As depicted in Fig. 6, to achieve this fusion, we modified the HR-Net by combining the feature map E generated by the texture enhancement module with the low-dimensional feature map from HR-Net after transition 1. Subsequently, a 1×1 convolution was applied to adjust the number of channels in the fused feature map to match that of the original low-dimensional feature map. The subsequent feature extraction process follows the overall workflow of HR-Net.

3.3 Multi-scale Feature Fusion

The multiscale feature fusion module is designed to enhance the model's performance by integrating feature maps at different scales, as shown in Fig. 7. The inputs to this module are the output features from the three branches, namely the texture feature D_tm combined with RGB features, the frequency-domain feature D_{fn}, and the noise feature D_{ni}, where $m \in \{1, 2, 3, 4\}$, $n \in \{2, 3, 4\}$ and $i \in \{1, 2, 3, 4\}$. Each input feature map is decomposed into a series of subfeature maps. It can be seen that the number of subfeature maps varies across different branches. The varying number of feature maps allows each branch to better represent the feature levels it is responsible for. This enables the model to better adapt to input data of different scales and

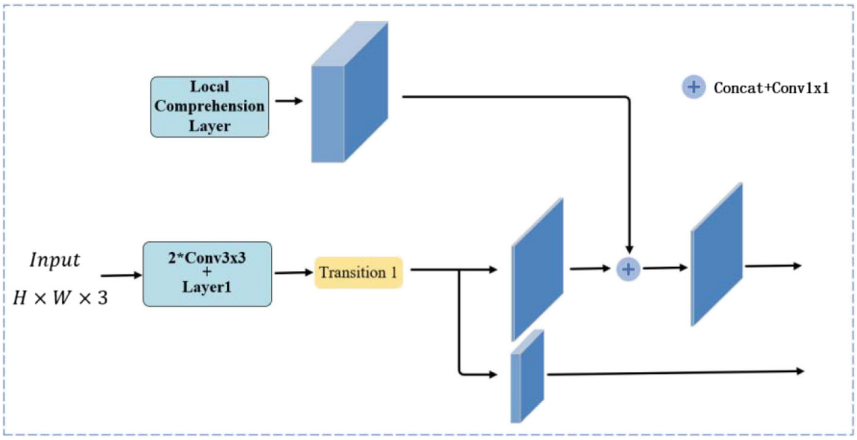


Fig. 6 We combine the regional attention layer with the backbone network so that the RGB features are fused with the texture features.

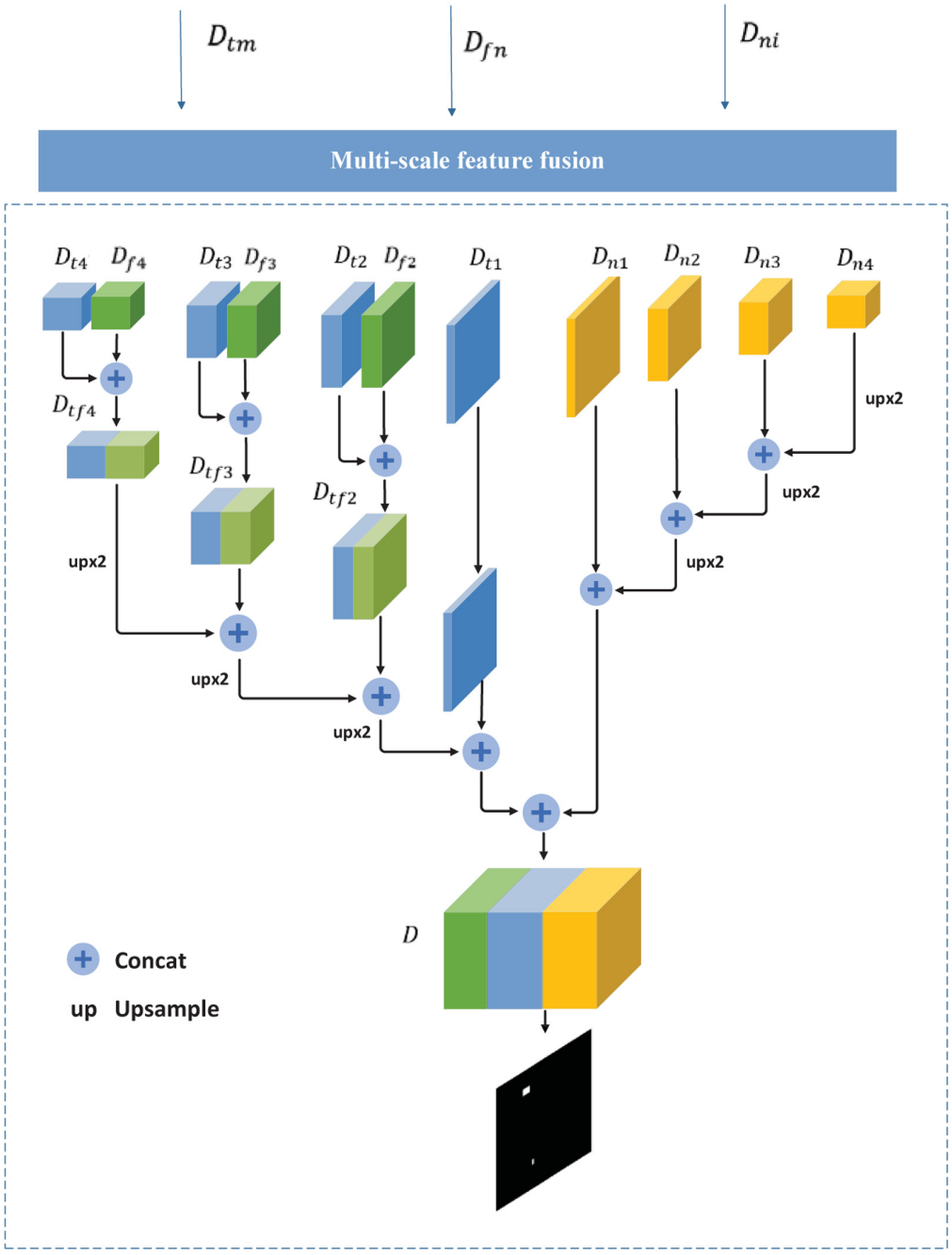


Fig. 7 Multiscale feature fusion module. Fully integrate multiview tampering features across different scales.

complexities. Each branch can be optimized for specific scales and features, thereby improving the overall performance.

In this module, the feature maps from different branches are upsampled and then passed down along the path, where they are combined with the feature maps from other branches. This cross-branch fusion helps to integrate features at different scales, thereby enhancing the model's ability to capture inputs of various scales. Finally, the feature maps are concatenated, merging all the fused feature maps into a single multichannel feature map. This step retains the information of all input features and provides rich context for the subsequent processing steps. For the representation of the entire process of the multiscale feature fusion module, we first define $U^k(\cdot)$ to denote the operation of performing k Upsample$(\cdot)$ operations. This process can be expressed as

$$U^k(x) = \text{UnSample}(U^{(k-1)}(x)). \tag{6}$$

Subsequently, the fusion of the noise feature branch is denoted as

$$D_n = D_{n1} + U^1(D_{n2} + U^1(D_{n3} + U^1(D_{n4}))). \tag{7}$$

Similarly, for the fusion of the frequency-domain feature branch and the texture feature branch, we first define

$$D_{tfi} = D_{ti} + D_{fi}, \tag{8}$$

where $i = 2,3,4$, and we similarly use the defined function $U^k(\cdot)$:

$$D_{tf} = D_{t1} + U^1(D_{tf2} + U^1(D_{tf3} + U^1(D_{tf4}))), \tag{9}$$

where D_n denotes the final fused features of the noise feature branch and D_{tf} denotes the final fused features of the texture and frequency-domain feature branches. Finally

$$D = D_n + D_{tf}. \tag{10}$$

In the aforementioned process, all "+" symbols represent the concatenate operation, and D denotes the final feature representation obtained after the features from all branches have passed through the multiscale feature fusion module.

4 Tamper-Identity Dataset

The training set currently consists of 900 complete tampered identity document images, covering the four mainstream tampering types: copy-paste, splicing, erasing, and generation. Both manual and automatic tampering methods were utilized. Figure 9 presents some image samples from the dataset. All textual content in the dataset is composed of Chinese characters and numbers, and it includes photos of portraits. Because the photos feature real portraits and involve privacy issues, privacy-protection measures have been applied to them.

The creation of our proposed tampering dataset Tamper-Identity comprises three steps, as illustrated in Fig. 8:

(1) The first step involves conducting initial editing on blank identity document templates with textures to fill in the content and create the original identity document images.
(2) The second step is to create authentic identity document image originals.
(3) The third step involves performing secondary editing on the identity document image originals to obtain tampered identity document images.

The training set currently consists of 900 complete tampered identity document images, encompassing the four mainstream tampering types: copy-paste, splicing, erasing, and generation. Both manual and automatic tampering methods were utilized. Figure 9 presents some image samples from the dataset. All textual content in the dataset is composed of Chinese characters and numbers, and it includes portrait photos. Because the photos feature real-life portraits and involve privacy concerns, privacy-protection measures have been applied to the photos.

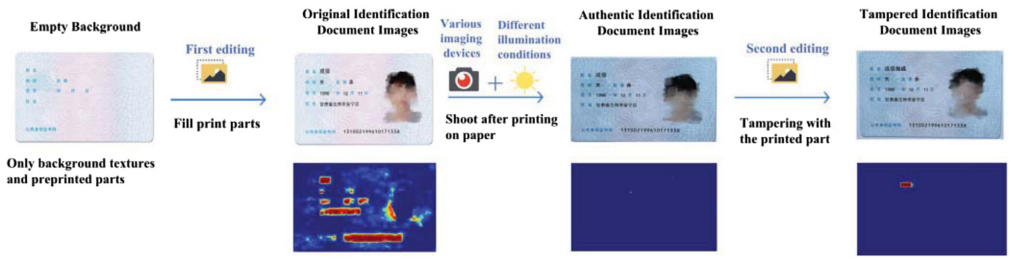


Fig. 8 Tamper-Identity creation process begins with editing blank identity document templates with background textures to create the original identity document images. After printing, the images are captured using different devices and lighting conditions to obtain authentic identity document image originals. Finally, the authentic images undergo tampering to produce tampered identity document images.

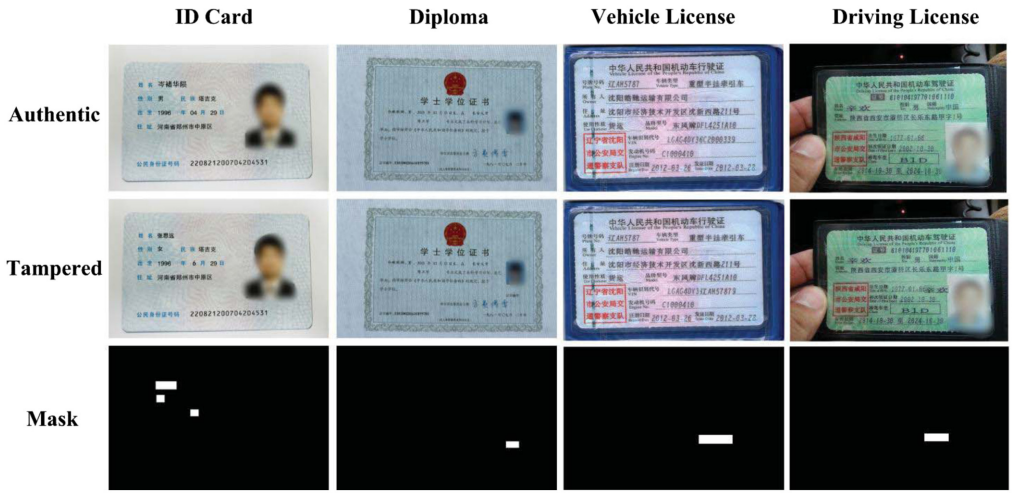


Fig. 9 Some image samples from the Tamper-Identity dataset, including ID cards, academic certificates, vehicle licenses, and driver's licenses, which are four common types of identity documents.

4.1 Creating the Original Identity Document Images (First Editing)

We collected blank identity document templates with textures (Fig. 10, with ID cards serving as an example, and the production process for other tampered identity document images is the same), i.e., images that only have background texture elements and preprinted parts (such as name, gender, major, and other text parts that will not be tampered with, see Fig. 11).

After obtaining the blank identity document templates, the printed parts and portrait images (all portrait images are sourced from the internet and have undergone privacy-protection treatment) are generated in the corresponding positions within the image. In this way, the original identity document images are created. It is important to note that the original identity document



Fig. 10 Blank document templates with background textures.



Fig. 11 Preprinted parts and printed parts (the tampering operations on document images generally focus on the printed parts).

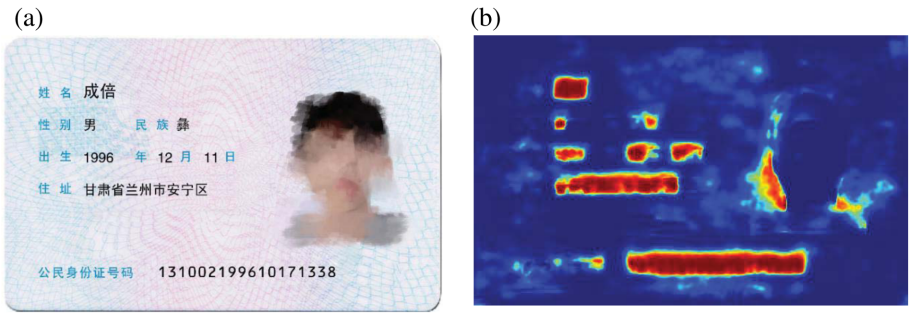


Fig. 12 Original identity document image after the first editing (a) and the visualization of tampering traces (b), where deeper heatmap colors indicate a higher likelihood of tampering.

images produced in this process have, in fact, undergone one editing step (the blank document template has been altered). Therefore, in reality, the original identity document images are already tampered images full of tampering traces. Figure 12 presents the original identity document images obtained after the first editing of the blank document template, along with the visualization of the tampering traces from this initial editing.

4.2 Creating Authentic Identity Document Images

In Sec. 4.1, we noted that the original identity document image obtained after one editing of the blank document template is already a tampered image, and thus all the printed parts are regarded as tampered regions, which obviously does not align with real-world conditions. Therefore, we utilized various shooting devices, such as mobile phones, tablets, and cameras, to photograph the original identity document images under diverse lighting conditions and then printed them. The resulting authentic identity document images do not bear tampering traces (see Fig. 13), becoming an untampered genuine image. Meanwhile, the different lighting conditions and shooting devices also add to the difficulty of subsequent tampering detection.

4.3 Creating Tampered Identity Document Images (Second Editing)

Tamper-Identity includes the tampering of authentic identity document image originals, covering both manual and automatic tampering methods. The former manually tampers with images using four mainstream methods in image editing software, aiming to achieve realistic forgeries. The latter automatically tampers with images through programs, creating tampered text by executing PS scripts that simulate the actual text tampering process, a common method for large-scale text tampering. This program introduces randomness to increase the diversity of Tamper-Identity. Specifically, OCR tools are used to first separate the text from the background, assign appropriate targets, and then perform the tampering operations.

Most existing tampered image datasets are created through a single-time editing process, which involves directly tampering with the original images. Tampered images created through a single-time editing may have tampering traces erased after image compression, which poses significant challenges for the tampering detection task. Malefactors can readily hide tampering traces by simply compressing the image once after tampering. After the first editing, Tamper-Identity uses different lighting conditions and shooting devices to simulate real identity

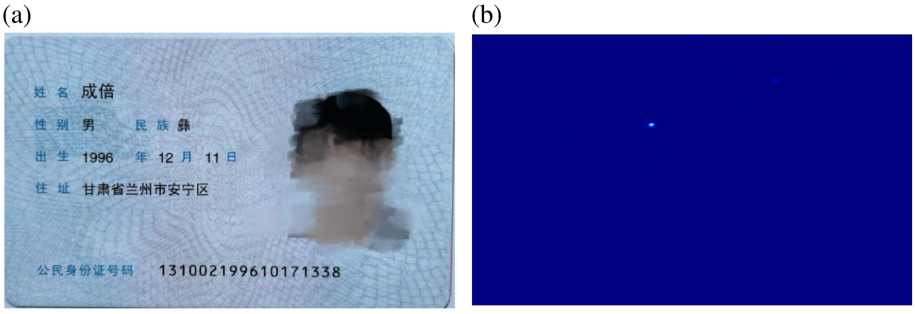


Fig. 13 Authentic identity document image (a) and the visualization of tampering traces (b).

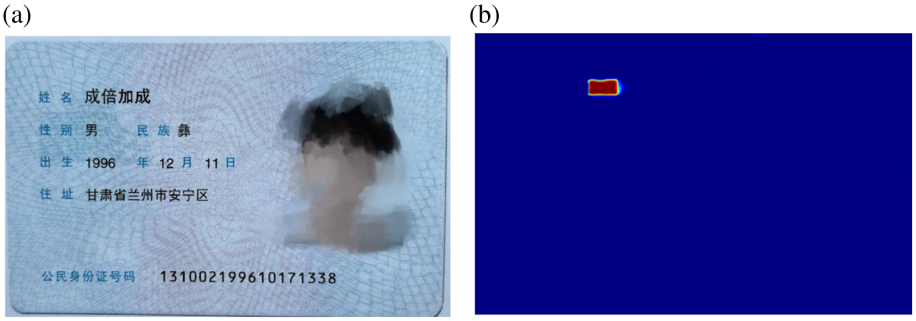


Fig. 14 Tampered identity document image (a) and the visualization of tampered regions (b).

document images and then undergoes a second editing to obtain the final tampered identity document images as shown in Fig. 14. Therefore, the Tamper-Identity dataset is of higher quality than most current datasets that have only been edited once. Different lighting conditions and shooting devices also contribute to enhancing the model's robustness and generalization capabilities.

5 Results and Discussions

5.1 Implementation Details

We deployed the network model on an RTX4090 GPU and optimized the network parameters using the stochastic gradient descent (SGD) method. During the training process, we utilized mixed-precision training to accelerate computations and reduce memory usage. The input image size was set to 512×512, and the model was trained using the training dataset from Tamper-Identity. The learning rate started at 0.005 and was exponentially decayed on a periodic basis. The batch size for training was set to 12, and the number of iterations was 150,000. In addition to tamper-identity, we also conducted experiments on the public datasets DocTamper and Tampered-SROIE.

5.2 Evaluation Metric

We employ IoU, recall, precision, and F-score as the evaluation metrics for tampering localization within our Tamper-Identity dataset. For other public datasets, we follow the previous work. The threshold of IoU is set to 0.5. The specific calculation methods are as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{11}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{12}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FN} + \text{FP}} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}, \tag{13}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{14}$$

In the evaluation system for image tampering detection, the following metrics have significant advantages: true positives (TP), which signify the number of pixels accurately identified as tampered; false positives (FP), which denote the number of untampered pixels incorrectly identified as tampered; and false negatives (FN), which represent the number of tampered pixels incorrectly classified as un-tampered. IOU, recall, precision, and F-score can effectively reflect the detection performance of text tampering regions, providing a comprehensive evaluation of the text tampering detection effect.

5.3 Comparison Experiment

To comprehensively evaluate the performance of the proposed identity document image tampering detection model based on texture understanding and a multistream network, we designed a series of rigorous comparative experiments. The experiments selected four advanced models

that are representative in the field of image forensics in recent years as benchmark methods: Mantra–Net,[24] CAT–Net,[19] MVSS–Net,[11] and RRU–Net.[25] These comparative methods cover the evolution from traditional feature analysis to deep learning and can fully reflect the current technical development level in the field of tampering detection.

The experiments employed the self-created Tamper-Identity identity document tampering dataset, which includes 900 high-quality identity document image tampering samples, each with pixel-level annotated tampering masks. To ensure the fairness of the experiments, we strictly followed the standard data division principle and randomly divided the dataset into a training set and a testing set in an 8:2 ratio. During the data preprocessing stage, we implemented data augmentation strategies for the training data, including random horizontal/vertical flipping, rotation by ± 15 deg, brightness adjustment ($\pm 20\%$), and Gaussian noise injection ($\mathcal{O} = 0.01$), to enhance the model's generalization ability. The testing set, on the other hand, maintained the original data distribution and was not subjected to any augmentation.

Table 1 presents the quantitative evaluation results of each model on the testing set. The proposed model in this paper demonstrates significant advantages on key metrics: the precision reaches 0.882, which is a 27.6% improvement over the second-best model; the recall is 0.941, a 26.7% increase compared to the second-best model; and the F-score reaches an excellent level of 0.912, a 27.7% improvement over the second-best model. Notably, in terms of the false negative rate, which is highly sensitive in identity document tampering detection, our model maintains a low misjudgment probability of 0.059. These results fully demonstrate the effectiveness of the multistream network architecture in fusing global semantics with local texture features, as well as the texture understanding module's precise capturing ability of the distinctive background textures in identity document images.

As shown in Table 1, the precision, recall, and F-score all exhibit a trend of gradual improvement, indicating that the detection performance of the model is continuously enhancing. Specifically, the early method Mantra-Net performs poorly in terms of precision and recall, and the F-score is also low, indicating significant limitations in detecting and localizing tampered regions. With the advancement of technology, RRU-Net and MVSS-Net have shown improvements in precision and recall, but there is still room for further enhancement. CAT-Net further elevates the precision and recall, particularly achieving high levels in precision and the F-score, demonstrating superior detection performance. Although these methods perform well on general datasets, the images in Tamper-Identity, which are created through secondary editing, interfere with the detection capabilities of the aforementioned methods, resulting in a decline in their performance. In addition to conducting comparative experiments using the self-made dataset Tamper-Identity, this paper also performed experiments on the large-scale dataset DocTamper constructed by Qu et al.[26] The training set of the DocTamper dataset was used to train the method proposed in this paper, and the comparison was carried out on the testing set. Table 2 presents the comparative experimental results of different methods.

The method proposed in this paper achieved the highest recall rate of 0.683, indicating a significant advantage in the number of detected tampered regions. Meanwhile, our method also performed well in terms of the F-score (0.656), demonstrating that the method can not only

Table 1 Comparison experiments conducted on the self-made dataset Tamper-Identity.

	Tamper-Identity			
Method	*P*	*R*	*F*	IoU
Mantra-Net[24]	0.061	0.143	0.082	0.045
RRU-Net[25]	0.142	0.123	0.135	0.071
MVSS-Net[11]	0.305	0.414	0.352	0.213
CAT-Net[19]	0.606	0.674	0.635	0.469
Ours	0.882	0.941	0.912	0.836

Table 2 Comparison experiments conducted on the testing set of the public dataset DocTamper.

	DocTamper		
Methods	*P*	*R*	*F*
Mantra-Net[24]	0.123	0.204	0.153
MVSS-Net[11]	0.494	0.383	0.431
PSCC-Net[13]	0.309	0.506	0.384
BEIT-Uper[27]	0.564	0.451	0.501
Swin-Uper[28]	0.671	0.608	0.638
Ours	0.632	0.683	0.656

effectively detect more tampered regions but also maintain a high level of precision when detecting and localizing tampered regions. Overall, the method has excellent performance and good generalization ability. However, the precision of our method was only the second-best. A more in-depth examination of the outcomes has shown that, within the DocTamper images, there's an absence of uniquely identifiable background texture characteristics. This particular aspect has had the effect of somewhat toning down the detection capabilities of the method we have put forward. The lack of distinctiveness in background textures means there is less pronounced contrast and definition for the detection mechanism to latch onto, leading to a slight decline in the precision and effectiveness of the detection results that our method is able to deliver. Nevertheless, the superior performance of our method in terms of recall rate and F-score compared with most current mainstream methods still proves that our method has good generalization ability.

Aiming to address the issue of performance degradation of the proposed method on different datasets and to evaluate whether our model learns robust features or merely overfits to the stylistic patterns in the Tamper-Identity dataset, we conducted experiments on the Tampered-SROIE[2] dataset. This dataset exhibits distinct language and layout characteristics while lacking photographic traces (e.g., illumination variations, paper wrinkles) and background texture features. As shown in Table 3, our method demonstrates strong generalization capability and effectiveness in learning robust features, even in the absence of unique background textures.

Figure 15 further substantiates the quantitative analysis results through visual comparison. The visualization results illustrate that the proposed model significantly outperforms others in detecting and localizing tampered regions. Although other methods can identify some tampered regions in specific instances, their overall detection outcomes are relatively sparse. There are evident omissions in detecting subtle tampering regions, and the detected tampered regions considerably deviate from the actual ones. In some critical areas, the tampering traces are not accurately identified, indicating certain limitations in their detection capabilities and the need for improvement in precision and robustness.

Table 3 Comparison experiments conducted on the testing set of the public dataset Tampered-SROIE.

	Tampered-SROIE		
Methods	*P*	*R*	*F*
EAST[29]	0.896	0.919	0.908
ATRR[30]	0.925	0.947	0.936
Ours	0.959	0.964	0.961

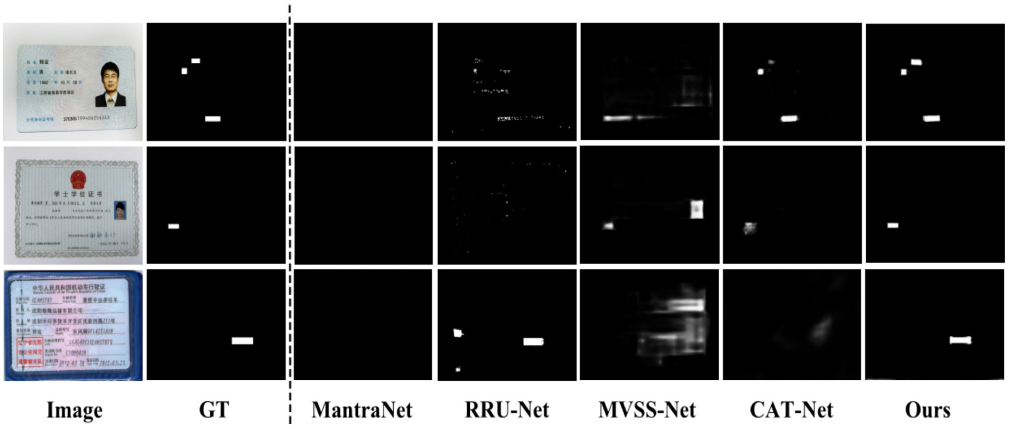


Fig. 15 Visualization of tampering localization results by different methods on Tamper-Identity.

By contrast, the proposed model demonstrates marked superiority in detecting and localizing tampered regions. The visualization results reveal that our model can accurately identify the majority of tampered regions and achieve extremely high localization accuracy. Notably, when dealing with complex scenarios and subtle tampering regions, our model can effectively capture the tampering traces, with the detection results highly consistent with the actual tampered regions. This indicates that our model has significant advantages in texture understanding and multistream network design, enabling it to more effectively address various challenges in identity document image tampering detection.

5.4 Ablation Analysis

To verify the effectiveness of each branch and components such as the texture enhancement module and the multiscale feature fusion module in our method, we conducted ablation experiments to evaluate the model's performance under different settings. The specific experimental setup was as follows: starting with the baseline model B_{rgb} that only had the RGB feature branch, we sequentially added the frequency-domain feature branch $B_{\text{frequency}}$, the noise feature branch B_{noise}, the texture enhancement module M_{texture}, and the multiscale feature fusion module (MSFF). We compared the tampering localization performance of these models on the testing dataset. The results are shown in Table 4. Our proposed complete model achieved the best performance, and the performance gradually improved as the key modules were added one by one. After adding the frequency-domain feature branch, the noise feature branch, and the texture enhancement module to form the complete multistream feature perceiver, the $F1$-score of the model on the testing dataset increased by ~23.1%, significantly enhancing the tampering localization ability. After incorporating the multiscale feature fusion attention module, the model paid more comprehensive attention to the tampering information in the image, resulting in higher-quality tampering localization mask maps.

Table 4 Ablation study conducted on the self-made dataset Tamper-Identity.

Methods					Tamper-Identity			
B_{rgb}	$B_{\text{frequency}}$	B_{noise}	B_{texture}	MSFF	P	R	F	IoU
√	—	—	—	—	0.167	0.294	0.212	0.119
√	√	—	—	—	0.606	0.674	0.635	0.469
√	√	√	—	—	0.565	0.793	0.660	0.492
√	√	√	√	—	0.869	0.914	0.891	0.803
√	√	√	√	√	0.882	0.941	0.912	0.836

Table 5 Comparison experiments on Tamper-Identity dataset with different compression quality. The IoU metric is used in all the experiments.

	Tamper-Identity		
Method	$Q = 90$	$Q = 85$	$Q = 80$
Mantra-Net[24]	0.036	0.027	0.026
RRU-Net[25]	0.062	0.043	0.042
MVSS-Net[11]	0.192	0.175	0.149
CAT-Net[19]	0.432	0.399	0.375
Ours	0.811	0.805	0.793

5.5 Robustness Experiment

The JPEG compression method, as a lossy compression technique, functions by reducing file size through the removal of image redundancy. In practical applications involving document image storage and transmission, JPEG compression is widely utilized. However, this process may lead to the attenuation or obscuration of noise traces in tampered images, which can negatively impact the performance of image tampering detection models. To assess the resilience of the proposed method against JPEG compression artifacts, experiments were carried out on the Tamper-Identity test set across various JPEG compression quality factors. As shown in Table 5, the proposed method demonstrates robustness to JPEG compression. Its detection results outperform those of other models. Following JPEG compression, although the IoU metric experiences a minor decrease, the method maintains effective tampering localization capabilities.

5.6 Comparison with Commercial Methods

To further validate the effectiveness of the proposed method and its broad prospects in practical applications, we compared it with the identity document image tampering detection technology of Intsig Information, which has been widely deployed in commercial settings. The image tampering detection technology of Hehe Information has been widely adopted in the field of identity document image tampering detection and has achieved good tampering detection results in the detection of important documents such as driver's licenses and vehicle licenses, demonstrating its reliability and practicality in real-world application scenarios.

In 2023, Intsig Information was awarded the championship in the "Text Tampering Detection" track at the International Conference on Document Analysis and Recognition (ICDAR) challenge. This achievement fully demonstrates its technical strength and leading position in the field of text tampering detection.

Subsequently, in the 2024 Global AI Attack and Defense Challenge, Intsig Information once again leveraged its outstanding technical capabilities to win the championship in the "Financial Scene Voucher Tampering Detection" track, further consolidating its leading position in the field of image tampering detection. In addition, in 2024, Intsig Information joined hands with the China Academy of Information and Communications Technology (CAICT) to release the "Technical Requirements for Text Image Tampering Detection Systems." The release of this important standard not only reflects Intsig Information's authoritative position in the industry but also provides important guidance and regulation for the development of text image tampering detection technology.

Given Intsig Information's remarkable achievements and leading position in the field of image tampering detection, we chose it as the object of comparison to intuitively analyze the differences in detection performance between our method and existing advanced technologies for various types of tampered images. For this purpose, we carefully selected a variety of tampered images from the Tamper-Identity testing dataset for visualization, aiming to more intuitively present the performance of different methods in the tampering detection task.



Fig. 16 Visualization of tampering localization results by different methods on Tamper-Identity.

We compared our method with Intsig Information's "PS Tampering Detection System" available on their official website. Figure 16 displays the comparison of tampering localization results on images with different types of tampering. As can be seen from the figure, Intsig Information's tampering detection technology currently mainly targets the "generation" type of tampering and achieves good detection results on images with this type of tampering. However, for other types of tampering, such as "erasing," "splicing," and "copy-move," the detection results of Intsig Information show significant limitations and are unable to accurately locate the tampered regions. By contrast, our proposed method performs well on various types of tampered images and can accurately locate different types of tampered regions, demonstrating its superior performance and generalization ability in handling complex tampering scenarios.

Specifically, in the "Erase and Generation" type of tampered images, our method can clearly identify the erased and regenerated regions, whereas Intsig Information failed to effectively detect these tampering traces. In the "Splice" type of tampered images, our method can accurately locate the spliced regions, showing high sensitivity to splicing tampering, whereas the detection results of Intsig Information have significant deviations. In the "Copy-Move" type of tampered images, our method also performs well, being able to precisely identify the copied and moved regions, whereas Intsig Information failed to accurately detect these tampering behaviors. In summary, the comparative experimental results with Intsig Information demonstrate that the proposed method exhibits significant advantages in detecting various types of tampered images. It not only accurately locates the tampered regions but also shows good generalization ability and robustness, providing a more effective and reliable solution for the field of identity document image tampering detection.

6 Conclusion

We propose an identity document image tampering detection method based on texture understanding and a multistream network. This method uses a multistream feature perceiver to combine multiple branches to extract texture feature information, RGB feature information, frequency-domain feature information, and noise feature information from images. It fully explores tampering features and significantly improves the accuracy of locating tampered regions in identity document images using a multiscale feature fusion module to fully integrate these features. In addition, addressing the issues in current datasets, we created the tampered identity document image dataset Tamper-Identity by using mainstream tampering types and combining secondary editing. We conducted extensive experiments on our self-built dataset Tamper-Identity, demonstrating that the proposed model has good tampering localization performance for identity document images.

In future research, we will focus on enhancing the performance and application scope of image tampering detection technology from various perspectives. In terms of algorithm optimization, we will further improve the model's generalization ability, enabling it to be more widely

applicable to different types of image tampering detection tasks. This encompasses not only common tampering methods like copy-paste and splicing but also AI-generated image tampering. Meanwhile, we will pay close attention to the tampering detection of portrait images in identity documents. With the advancement of AI-generated content (AIGC) technology, the forgery of portrait images has become more realistic and less detectable. Therefore, it is necessary to conduct tampering detection on portrait images in identity documents, leveraging specific features of identity documents such as lighting consistency and edge artifacts, to enhance the accuracy and reliability of detection. Moreover, integrating multimodal technology to boost the model's tampering detection performance and interpretability is also a key research direction for us. By combining multimodal large-language models with visual and linguistic comprehension capabilities, we cannot only assess the authenticity of images and generate masks for tampered regions but also provide detailed explanations of tampering clues. This approach not only elevates detection precision but also enhances the credibility and transparency of detection outcomes.

In summary, future research will concentrate on enhancing the generalization ability of tampering detection technology, improving detection accuracy for specific scenarios (such as portrait photos in identity documents), and boosting the interpretability of detection through multimodal technology. This will better address the increasingly complex issues of image tampering.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code and Data Availability

The data that support the findings of this article are not publicly available due to privacy of personal portrait rights. They can be requested from the author at 17606433021@163.com.

Author Contributions

L.W. contributed to conceptualization, formal analysis, and writing—review and editing; Z.L. contributed to funding acquisition, methodology development, supervision, and validation; W.Z. contributed to writing—original draft. All authors reviewed the manuscript.

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